Two Port Networks- Transmission-Parameter, h-Parameter	
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Aim:

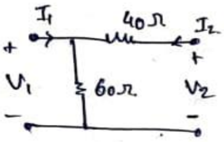
To verify the Two port Network Transmission Parameters and Hybrid parameters.

Theoretical Calculation:

Que 1. Calculate the h parameters of the given network.

Experiment – 7 Two port Network

Que 1 Calculate h parameter of the given Network



h parameters :

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$V_1 = 60(I_1 + I_2)$$

$$V_1 = 60I_1 + 60I_2 \quad \text{—①}$$

$$V_2 = 40I_2 + 60(I_1 + I_2)$$

$$V_2 = 40I_2 + 60I_1 + 60I_2$$

$$V_2 = 60I_1 + 100I_2 \quad \text{—②}$$

~~from ① $I_2 = \frac{V_1 - 60I_1}{60}$~~

$$I_2 = \frac{-60I_1 + V_2}{100} = \frac{V_2 - 60I_1}{100}$$

$$V_1 = 60I_1 + 60\left(\frac{V_2 - 60I_1}{100}\right)$$

$$V_1 = 60I_1 + \frac{6}{10}V_2 - 360I_1$$

$$V_1 = 24I_1 + \frac{6}{10}V_2$$

$$\boxed{V_1 = 24I_1 + 0.6V_2} \quad \text{—③}$$

$$V_2 = 60I_1 + 100I_2$$

$$I_2 = \frac{(60I_1 - V_2)}{100}$$

$$I_2 = \frac{60}{100}I_1 - \frac{V_2}{100}$$

$$\boxed{I_2 = 0.6I_1 - 0.01V_2} \quad \text{—④}$$

Puting I_2 in eq ②

$$V_2 = 60I_1 - 100\left(\frac{V_1 - 60I_1}{60}\right)$$

$$V_2 = 60I_1 - \frac{100V_1}{60} + 1000I_1$$

$$V_2 = 1060I_1 - \frac{100}{60}V_1$$

$$\frac{100}{6}V_1 = 1060I_1 - V_2$$

$$V_1 = \frac{1060 \times 6}{100}I_1 - \frac{6}{100}V_2$$

$$\cancel{V_1 = 63.6 - 0.06V_2}$$

$$V_1 =$$

$$I_1 = \frac{V_1 - 60I_2}{60} \quad \text{—(from ①)}$$

$$\cancel{V_2 = 60I_1 + 1 \cdot}$$

0.01 , 0.6

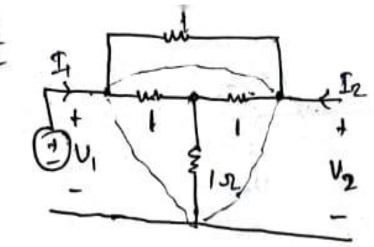
from eq ③ and ④

$$[h] = \begin{bmatrix} 24\Omega & 0.6 \\ 0.6 & 0.01S \end{bmatrix}$$

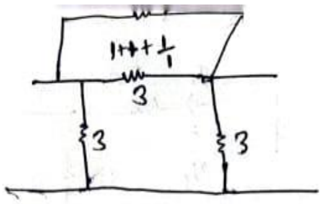
$$[h] = \begin{bmatrix} 24\,\Omega & 0.6 \\ 0.6 & 0.01\,S \end{bmatrix}$$

Que 2. Calculate the h parameters of the given network.

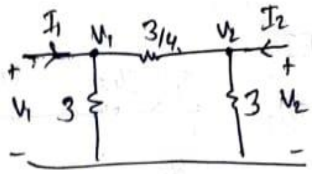
Que.2



$\longleftrightarrow$



$\frac{1\times 3}{4}$



$Y \longrightarrow \Delta \Rightarrow R_1 + R_2 + \frac{R_1 R_2}{R_3}$

KCL at V_1

$I_1 = \frac{V_1}{3} + \frac{V_1 - V_2}{3/4}$

$I_1 = \frac{V_1}{3} + \frac{4V_1 - 4V_2}{3}$

$I_1 = \frac{5V_1}{3} - \frac{4V_2}{3}$

$\frac{5V_1}{3} = I_1 + \frac{4}{3}V_2$

$V_1 = \frac{3}{5}I_1 + \frac{3}{5}\times\frac{4}{3}V_2$

$\boxed{V_1 = \frac{3}{5}I_1 + \frac{4}{5}V_2}$

KCL

$I_2 = \frac{5}{3}V_2 - \frac{4}{5}I_1 + \frac{16}{15}V_2$

$I_2 = -\frac{4}{5}I_1 + \frac{5\cancel{+16}}{15}V_2$

$\boxed{I_2 = -\frac{4}{5}I_1 + \frac{3}{5}V_2}$

$h = \begin{bmatrix} 0.6\Omega & 0.8 \\ -0.8 & 0.6S \end{bmatrix}$

For h Parameter

$V_1 = h_{11}I_1 + h_{12}V_2$

$I_2 = h_{21}I_1 + h_{22}V_2$

KCL at V_2

$I_2 = \frac{V_2}{3} + \frac{(V_2 - V_1)4}{3}$

$I_2 = \frac{V_2 + 4V_2 - 4V_1}{3}$

$I_2 = \frac{5}{3}V_2 - \frac{4}{3}V_1$

$I_2 = \frac{5}{3}V_2 - \left(\frac{3}{5}I_1 + \frac{4}{5}V_2\right)\frac{4}{3}$

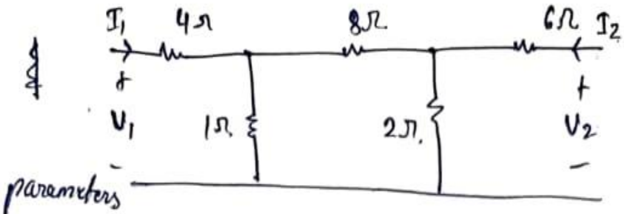
$I_2 = \frac{5}{3}V_2 - \frac{3}{5}I_1 - \frac{4}{5}V_2$

$I_2 = \left(\frac{5}{3} - \frac{4}{5}\right)V_2 - \frac{3}{5}I_1$

$\frac{20-12}{15} =$

Que 3. Calculate the T parameters of the given network.

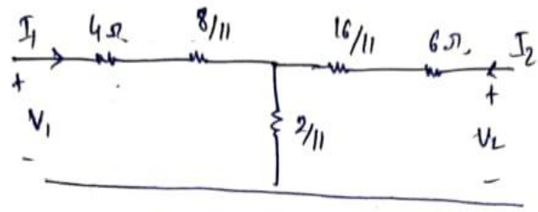
Ques



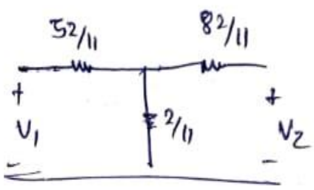
Transmission parameters

$V_1 = AV_2 - BI_2$

$I_2 = CV_2 - DI_2$



$$V_2 = \frac{84}{11} I_2 + \frac{2I_1}{11}$$



$$\frac{2}{11} I_1 = V_2 - \frac{84}{11} I_2$$

$$\boxed{I_1 = \frac{11}{2} V_2 - 42\, I_2}$$

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$V_1 = \frac{54}{11} I_1 + \frac{2}{11} I_2$$

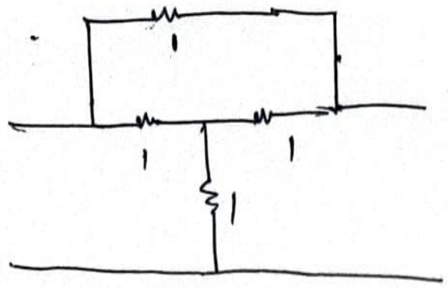
$$V_1 = \frac{54}{11}\left(\frac{11}{2} V_2 - 42 I_2\right) + \frac{2}{11} I_2 \qquad [T] = \begin{bmatrix} 27 & -206\,\Omega \\ 5.5\,℧ & -42 \end{bmatrix}$$

$$\boxed{V_1 = 27 V_2 - 206\, I_2}$$

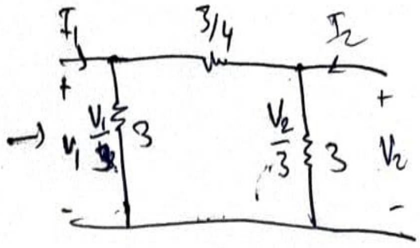
$$\boxed{I_1 = 5.5\, V_2 - 42\, I_2}$$

Que 4. Calculate the T parameters of the given network.

Que4



$\longrightarrow$



$$\Rightarrow \frac{3}{4}\left(I_2 - \frac{V_2}{3}\right) + V_1 = V_2 \qquad I_1 = -\left(I_2 - \frac{V_2}{3}\right) + \frac{V_1}{3}$$

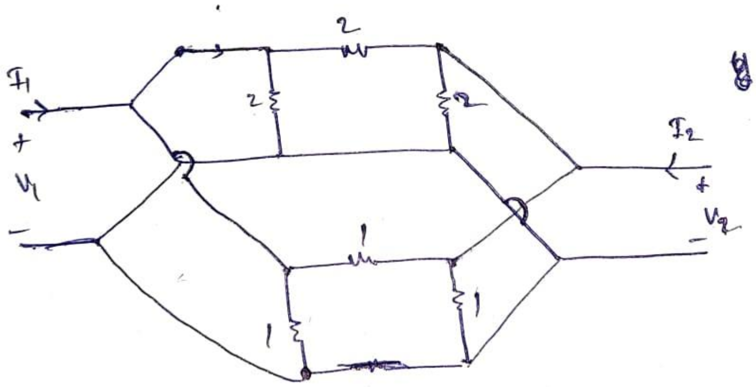
$$\Rightarrow V_1 = V_2 - \frac{3}{4} I_2 + \frac{1}{4} V_2 \qquad I_1 = -(I_2) + \frac{V_2}{3} + 1.25 V_2 - 0.25 I_2$$

$$\boxed{V_1 = 1.25 V_2 - 0.75 I_2} \qquad \boxed{I_1 = \frac{3}{4} V_2 - \frac{5}{4} I_2}$$

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \Rightarrow [T] = \begin{bmatrix} 1.25 & 0.75\,\Omega \\ 0.75 & 1.25 \end{bmatrix}$$

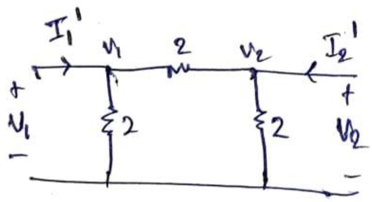
Que 5. Calculate the h parameters of the given network.

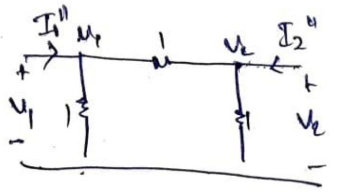
Ques



$I_1 = y_{11} V_1 + y_{12} V_2$

$I_2 = y_{21} V_1 + y_{22} V_2$





$I_1' = \frac{V_1 - V_2}{2} + \frac{V_1}{2}$

$I_1' = \frac{V_1 - V_2 + V_1}{2}$

$I_1' = \frac{2V_1}{2} - \frac{V_2}{2}$

$I_1' = V_1 - \frac{1}{2} V_2$

$I_2' = \frac{V_2}{2} + \frac{V_2 - V_1}{2}$

$I_2' = V_2 - \frac{1}{2} V_1$

$I_2' = -\frac{1}{2} V_1 + V_2$

$[y_a] = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$

$$[y_b] = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$I_1'' = \frac{V_1}{1} + \frac{V_1 - V_2}{1}$$

$$I_1'' = V_1 + V_1 - V_2$$

$$\boxed{I_1'' = 2V_1 - V_2}$$

$$I_2'' = \frac{V_2}{1} + \frac{V_2 - V_1}{1}$$

$$\boxed{I_2'' = -V_1 + 2V_2}$$

$$[y] = [y_a] + [y_b]$$

$$= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -3/2 \\ -3/2 & 3 \end{bmatrix} = [y]$$

$$[y] \longrightarrow [z]$$

$$[y] = [z]^{-1}$$

$$z = \begin{bmatrix} \overset{4/9=}{0.444} & 2/9 \\ \frac{2}{9} & 4/9 \end{bmatrix}$$

$$|z| = 0.148$$

$$h_{11} = \frac{\Delta z}{z_{22}} = \frac{0.148}{4/9}$$

$$h_{11} = 0.333$$

$$h_{12} = \frac{z_{12}}{z_{22}} = \frac{2}{9} \times \frac{9}{4} = \frac{2}{4} = 0.5$$

$$h_{21} = \frac{-z_{21}}{z_{22}} = -\frac{2}{9} \times \frac{9}{4} = -\frac{2}{4} = -0.5$$

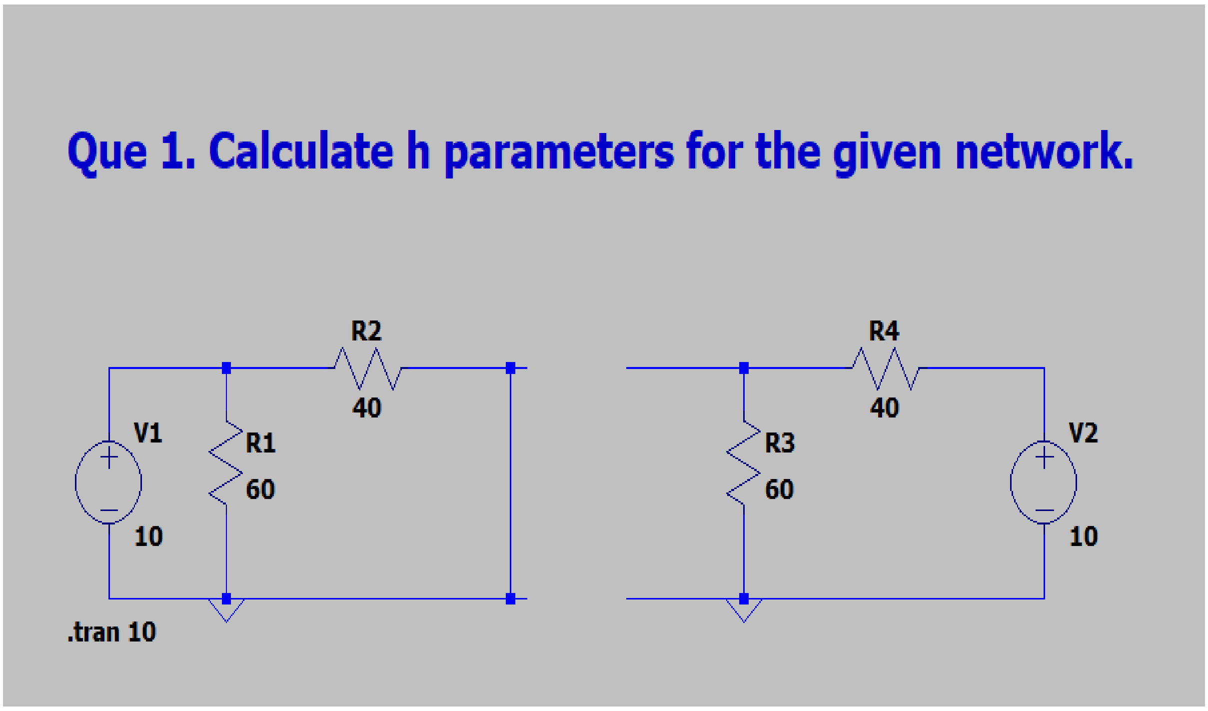
$$h_{22} = \frac{1}{z_{22}} = \frac{1}{z_{22}} = \frac{9}{4} = 2.25$$

$$[h] = \begin{bmatrix} 0.333\Omega & 0.5 \\ -0.5 & 2.25\,S \end{bmatrix}$$

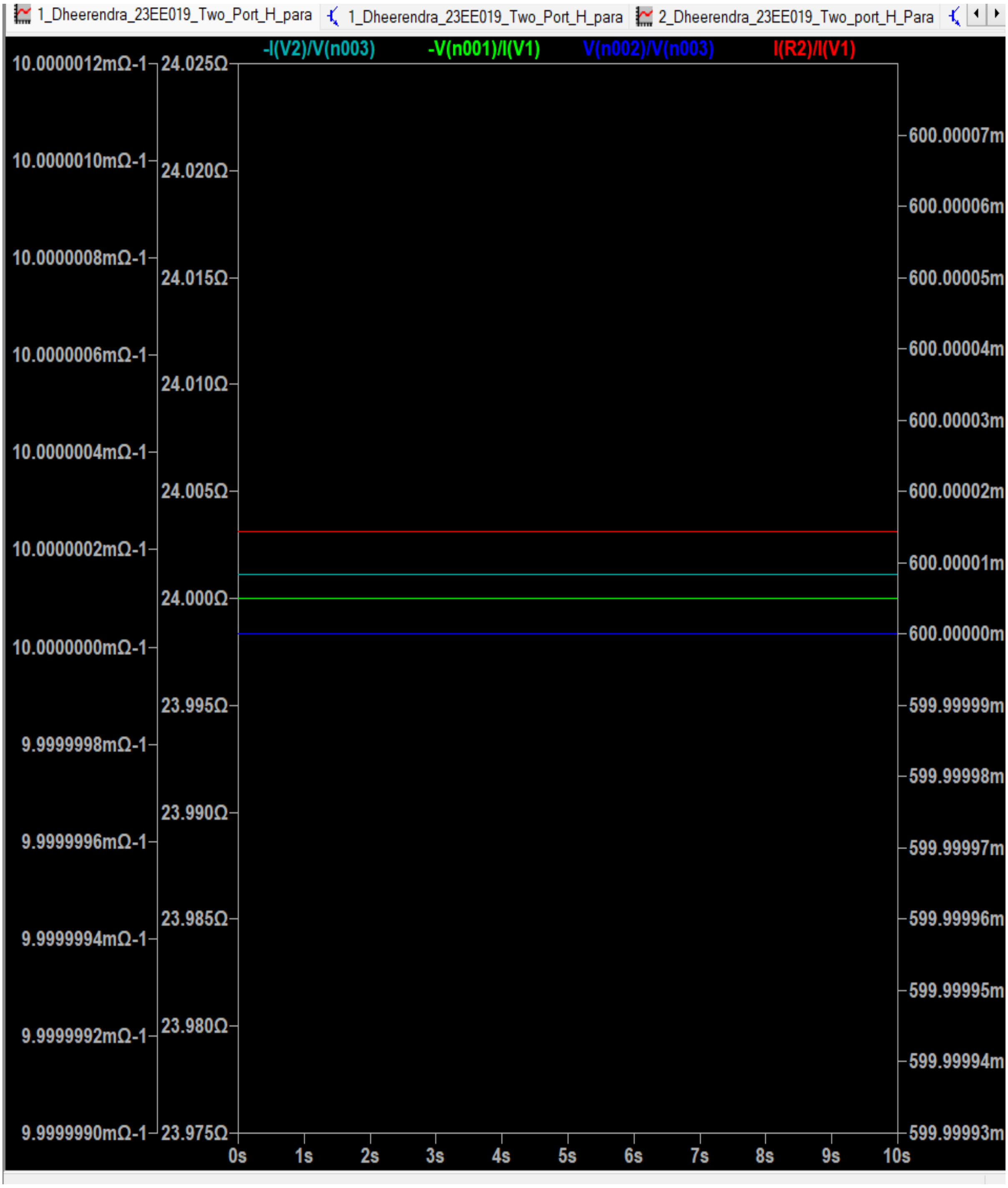
Verification by Simulation:

Que 1. Calculate h parameters for the given network.

Circuit Diagram:



Plot of Voltage/Current:

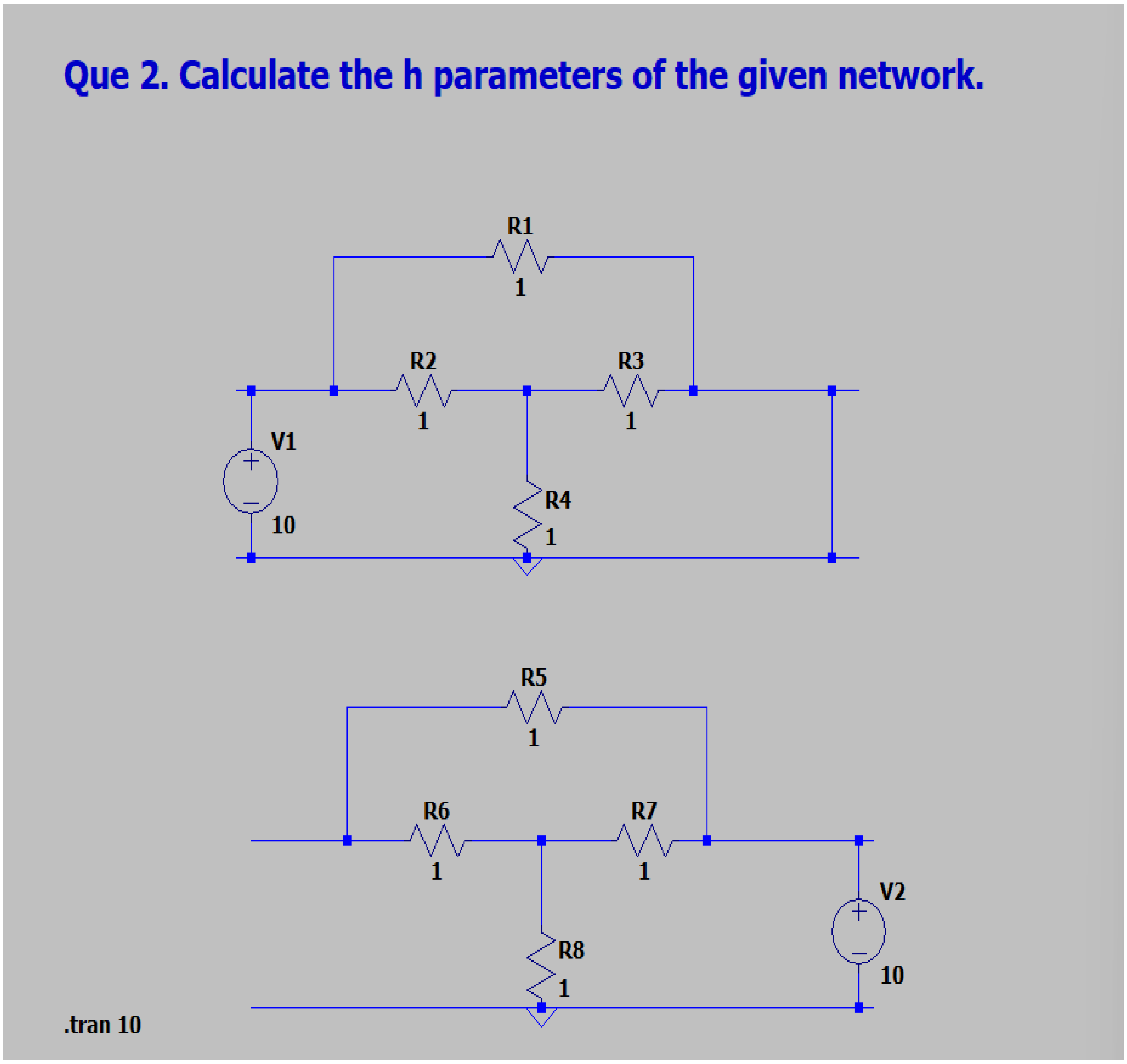


Observation Table:

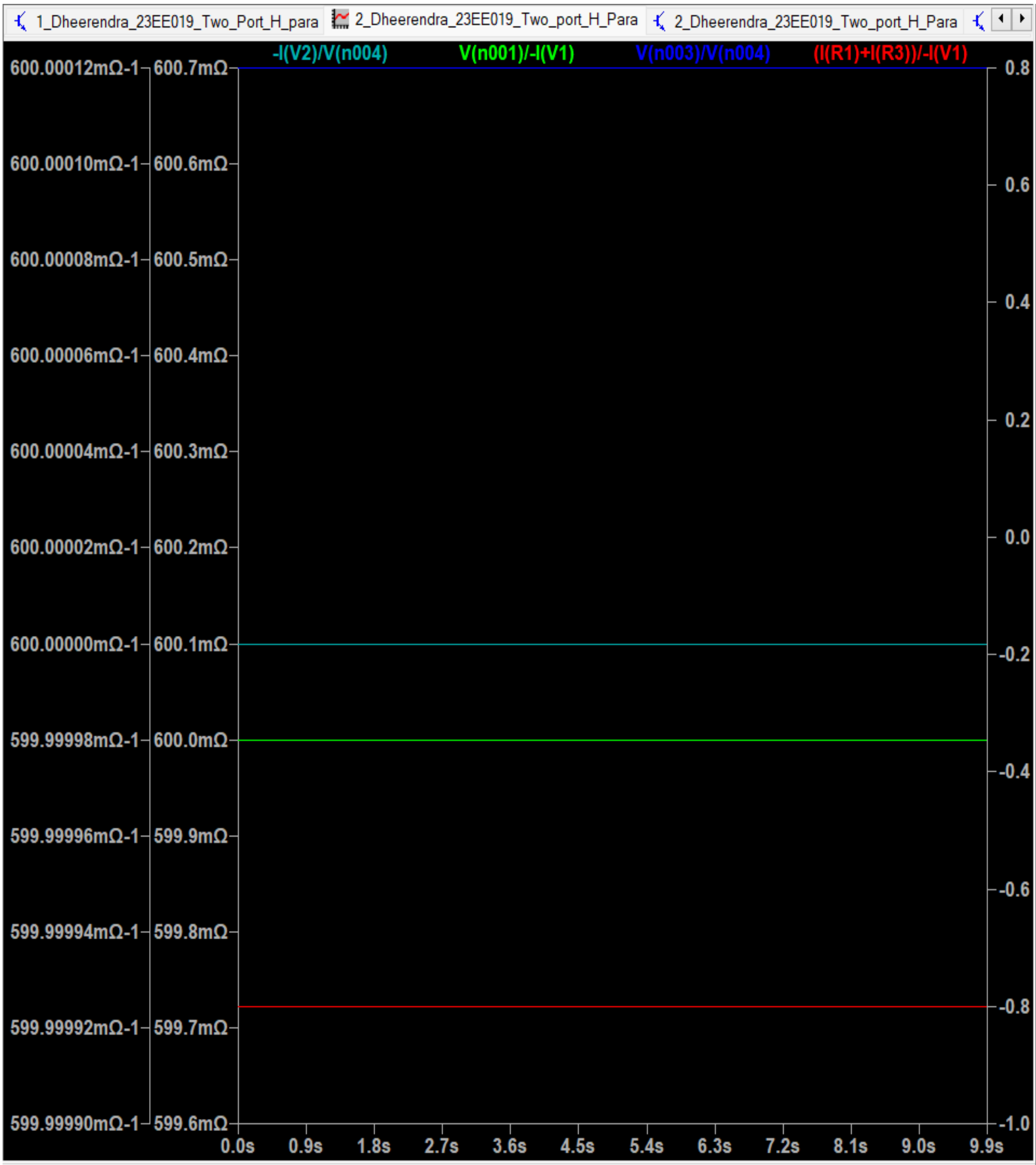
S. No.	h-Parameters	Theoretical Value	Practical Value
1	h_{11}	24 Ohm	24.000001Ohm
2	h_{12}	0.6	0.6
3	h_{21}	0.6	0.6
4	h_{22}	0.01 Ohm-1	0.01 Ohm-1

Que 2. Calculate the h parameters of the given network.

Circuit Diagram:



Plot of Voltage/Current:

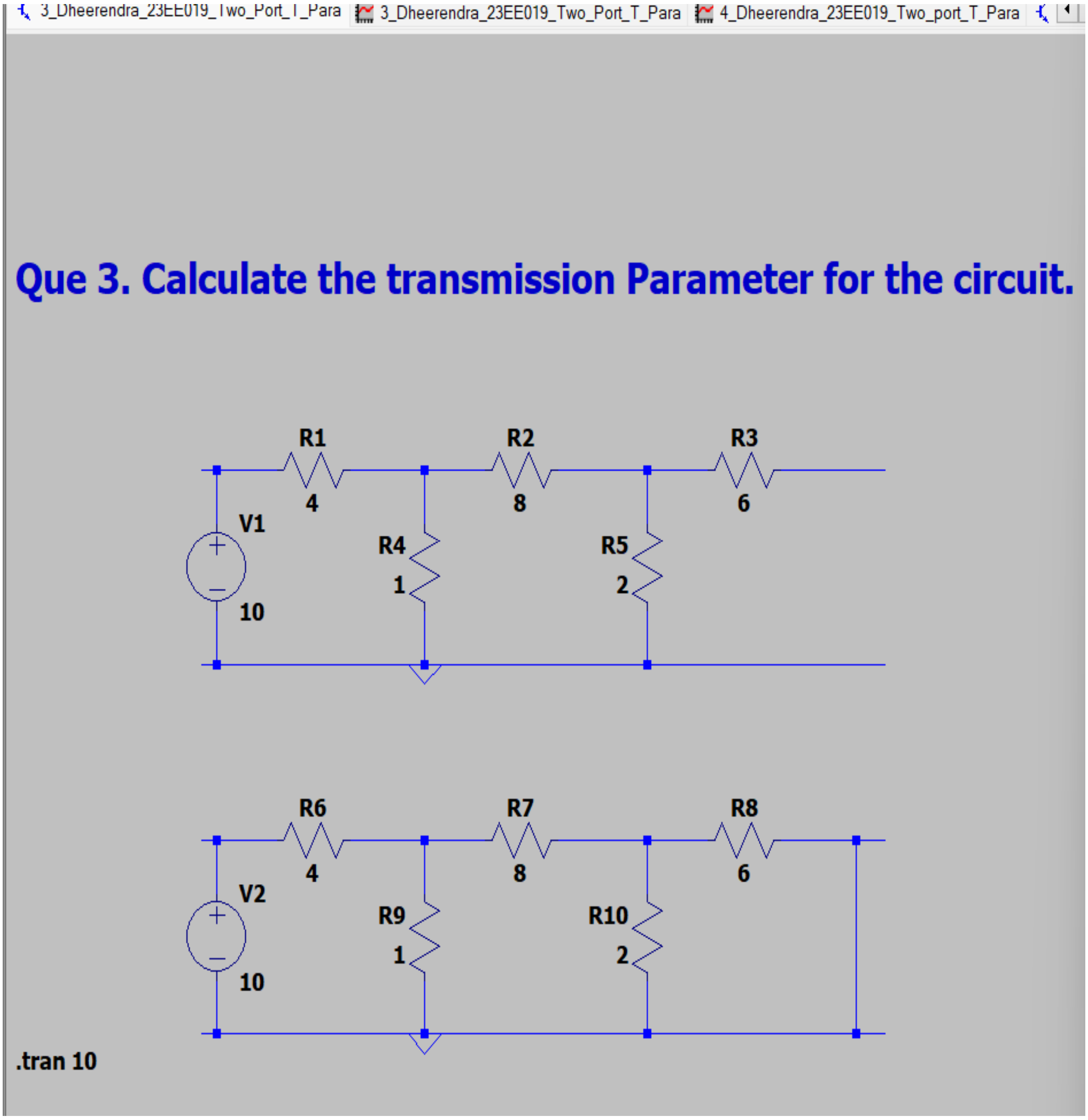


Observation Table:

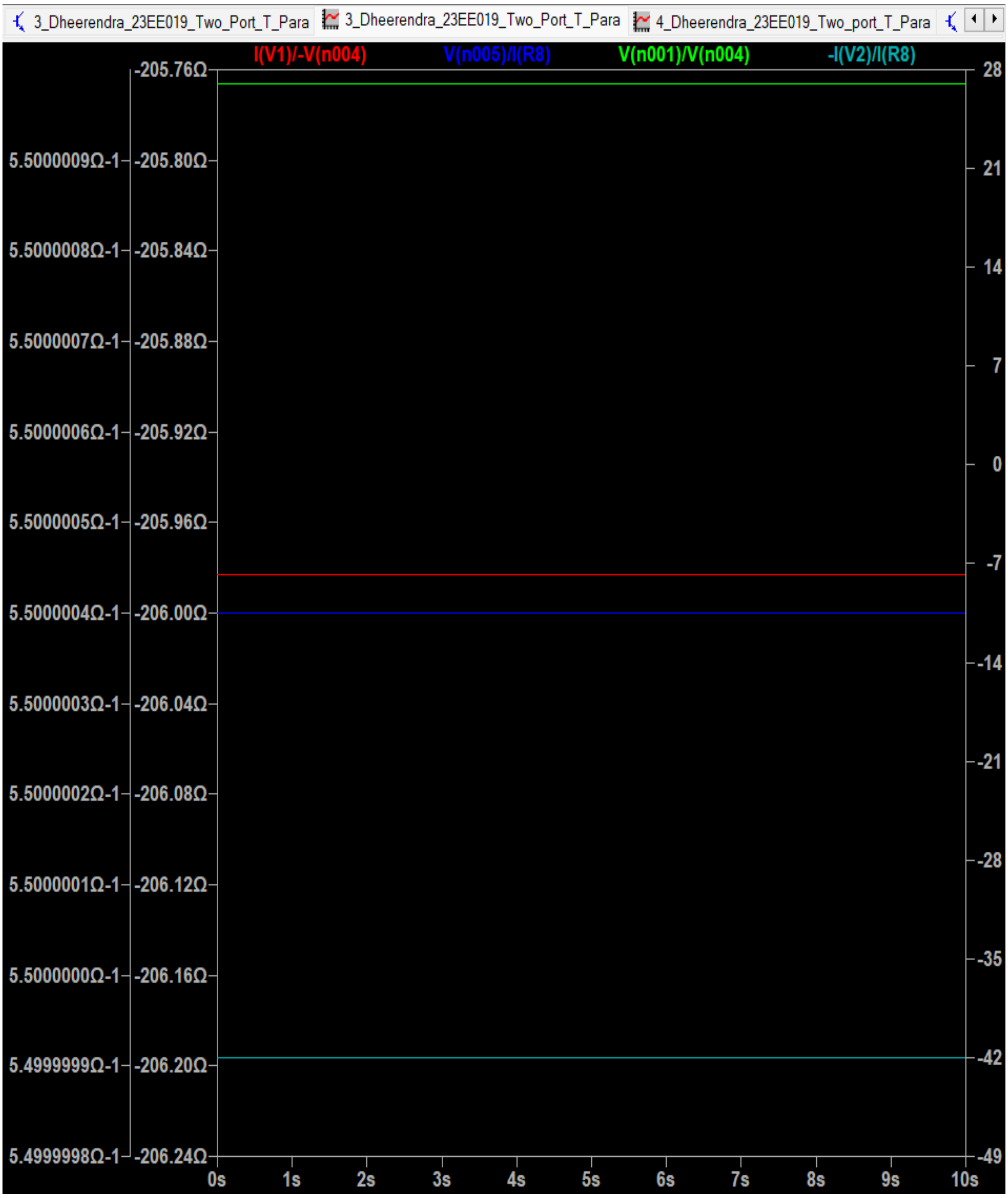
S. No.	h-Parameters	Theoretical Value	Practical Value
1	h_{11}	0.6 Ohm	0.6 Ohm
2	h_{12}	0.8	0.8
3	h_{21}	-0.8	-0.8
4	h_{22}	0.6 Ohm-1	0.6 Ohm-1

Que 3. Calculate the transmission Parameter for the circuit.

Circuit Diagram:



Plot of Voltage/Current:

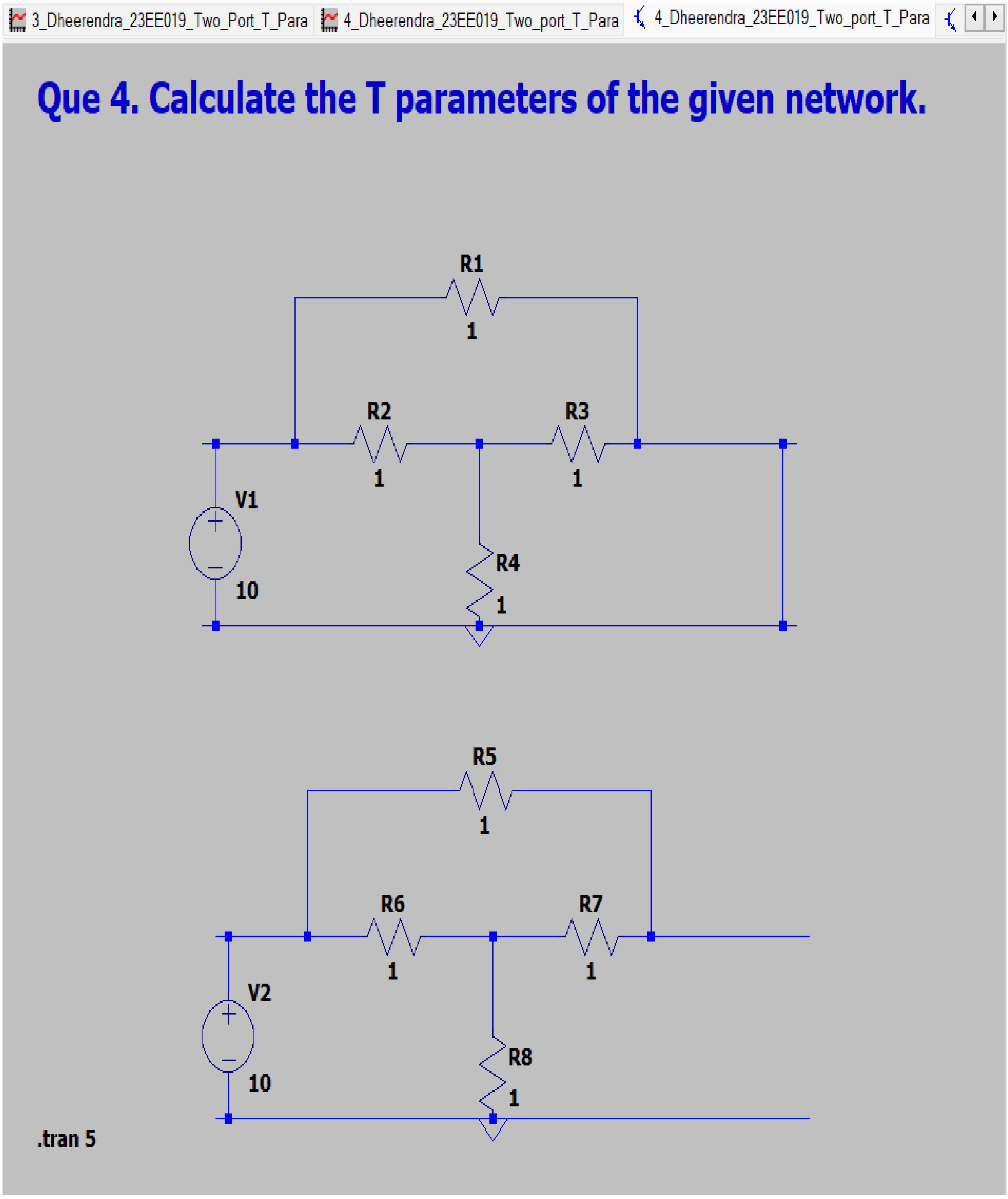


Observation Table:

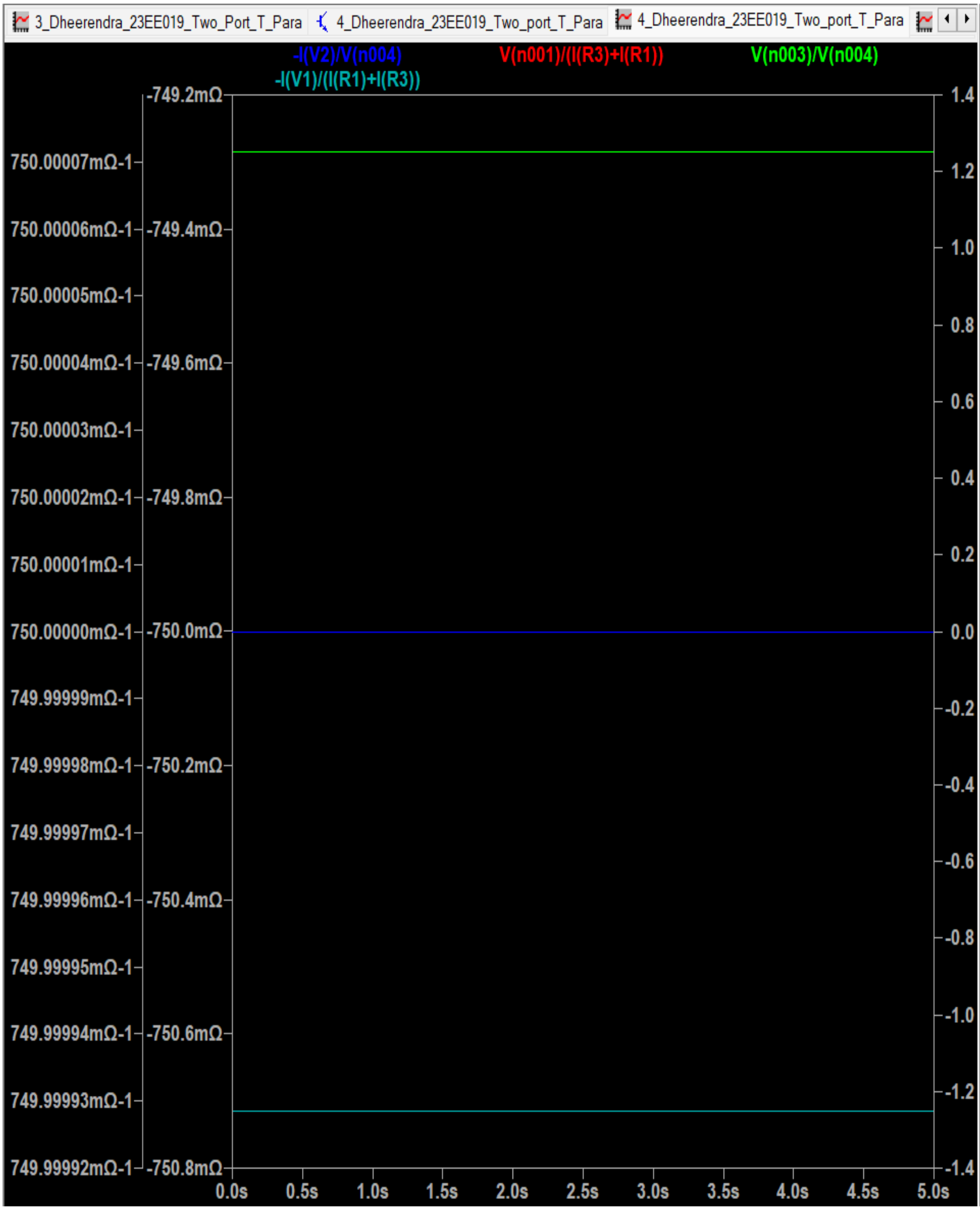
S. No.	T-Parameters	Theoretical Value	Practical Value
1	A	27	27.000001
2	B	-206 Ohm	-206.00001Ohm
3	C	5.5 Ohm-1	5.5000004Ohm-1
4	D	-42	-42.000003

Que 4. Calculate the T parameters of the given network.

Circuit Diagram:



Plot of Voltage/Current:

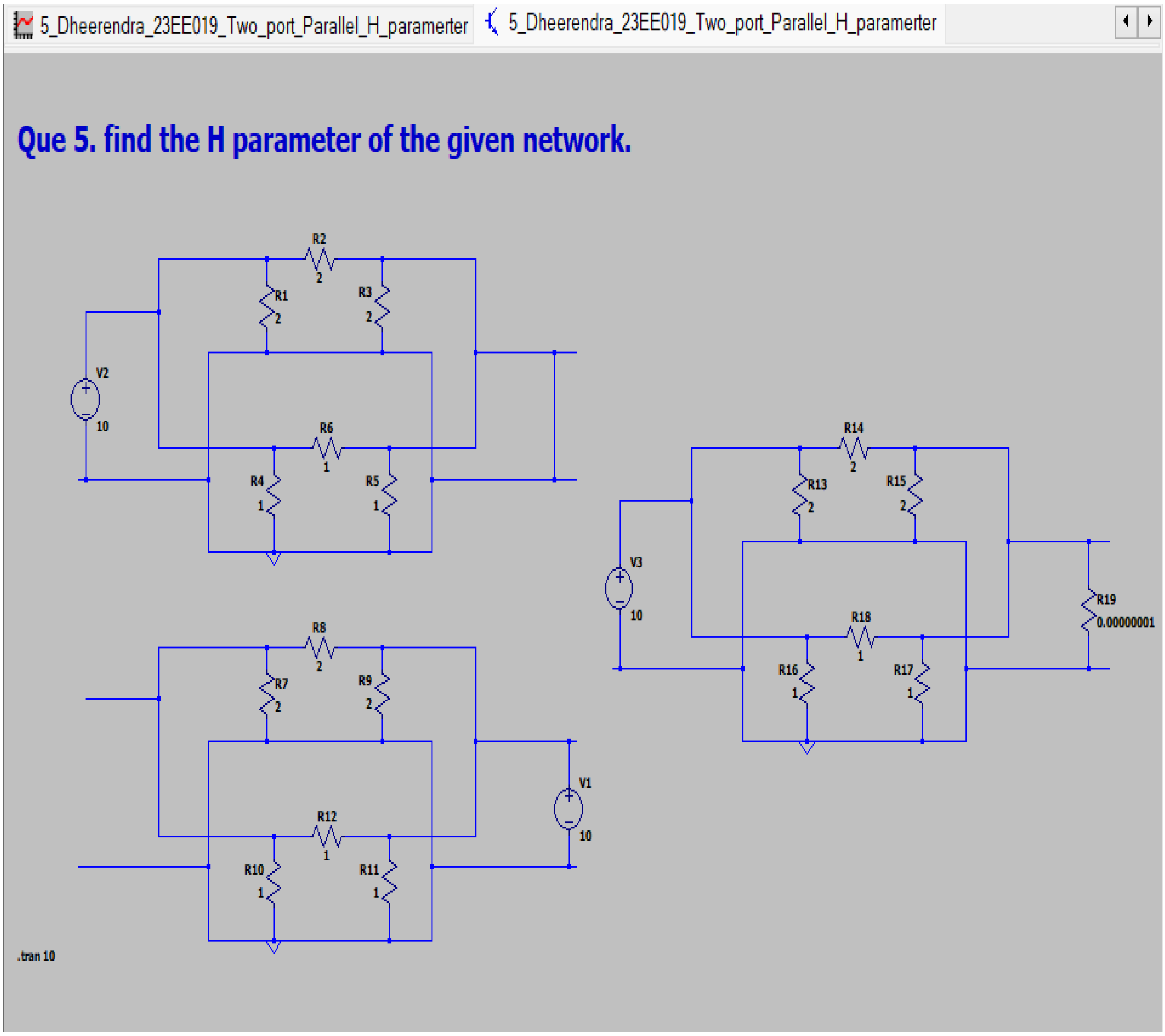


Observation Table:

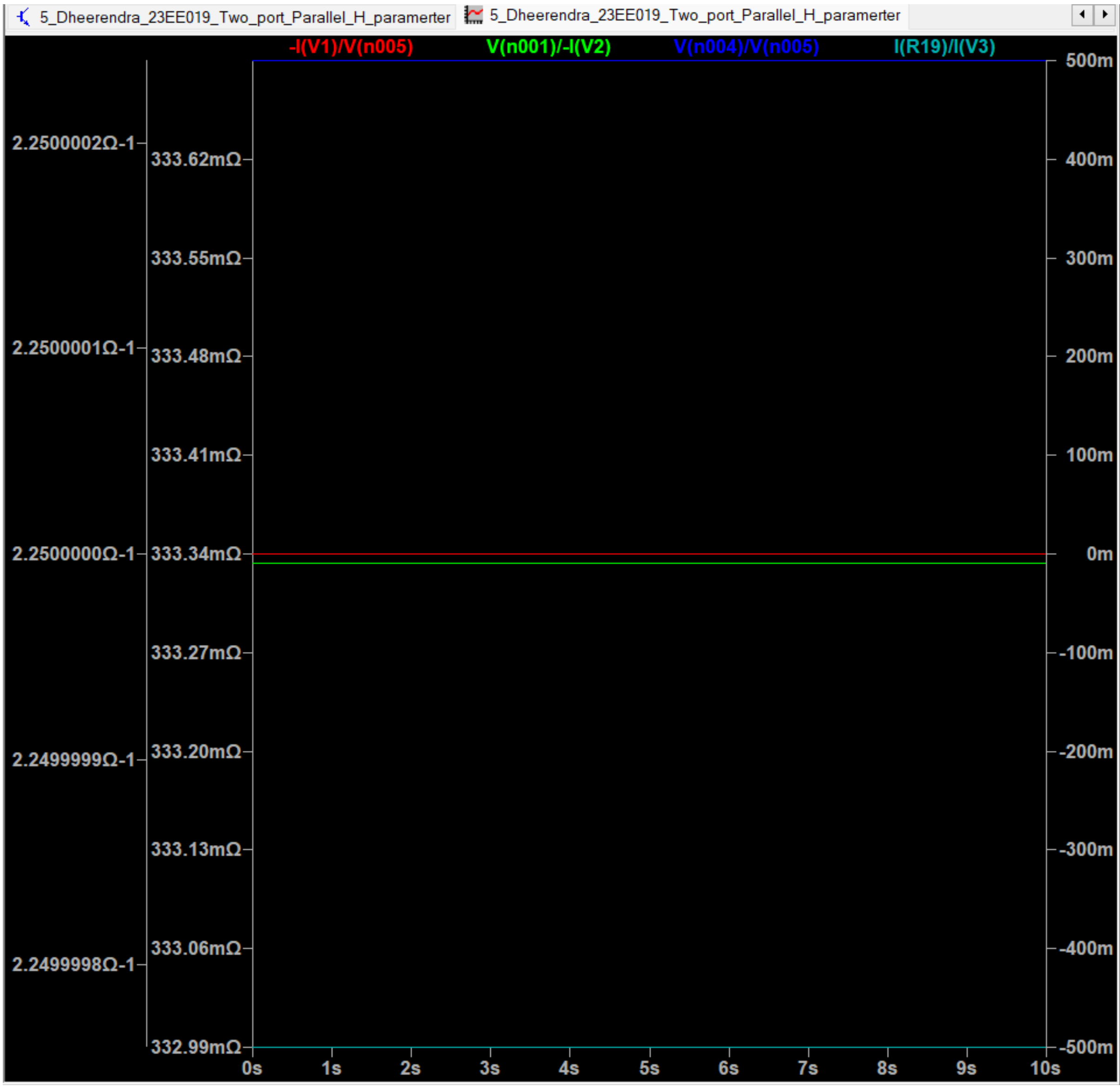
S. No.	T-Parameters	Theoretical Value	Practical Value
1	A	1.25	1.25
2	B	0.75 Ohm	750mOhm-1
3	C	0.75 Ohm-1	750mOhm
4	D	1.25	1.25

Que 5. find the H parameter of the given network.

Circuit Diagram:



Plot of Voltage/Current:



Observation Table:

S. No.	T-Parameters	Theoretical Value	Practical Value
1	A	0.333 Ohm	0.333333 Ohm
2	B	0.5	0.5
3	C	-0.5	-0.5
4	D	2.25 Ohm-1	2.25 Ohm-1

Results and Conclusion:

As we see that the theoretical values of the Transmission parameters and the Hybrid parameters is approximately equal to the practical values of the Transmission parameters and the Hybrid parameters, Hence the Z parameters and Y parameters are Verified.